

CHIRANJEEVI G

Security Engineer & Backend Lead

Promoted to Lead Security Engineer effective October 2026

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Chiranjeevi ORCID

PROFESSIONAL SUMMARY

Lead Security Engineer with 3 years across cloud security, DevSecOps, and secure architecture. Owns the security architecture and the CI/CD security gates for a multi-tenant managed security platform that runs on isolated per-client AWS deployments. Writes the production backends he secures, including a FastAPI, Supabase, Redis and AWS application backend shipped as sole developer, which most cloud security engineers cannot do. Seven published papers in cybersecurity and machine learning, security blogs past 100,000 cumulative reads, CEH v12 and CND certified.

SKILLS

- **Cloud security:** AWS, Azure, multi-account isolation, IAM boundaries, Cloudflare security and DNS, threat modeling, cost-aware architecture.
- **DevSecOps & CI/CD security:** pipeline security gates, Semgrep, Gitleaks, Trivy, Syft, Checkov, SBOM generation, supply-chain security.
- **Infrastructure as code & containers:** Terraform, Docker, Docker Compose, Kubernetes, Kubescape.
- **Detection & response tooling:** Wazuh, SIEM and EDR integration, DefectDojo, JIRA-integrated findings workflow.
- **Vulnerability management:** EPSS, CISA KEV, SSVC prioritization, SARIF normalization, triage automation.
- **AI security & agentic systems:** securing agentic pipelines, LLM-assisted triage and enrichment, agent and prompt threat surface, OpenAI, Llama, Ollama.
- **Programming & automation:** Python, Bash, FastAPI, REST API design, E2EE backend development.
- **Compliance & risk:** SOC 2 readiness, control mapping, audit readiness, regulated-client delivery.
- **Offensive security (background):** web and API penetration testing, white-box and black-box methodology, exploitation, attack surface analysis, network VAPT.
- **Self-hosted infrastructure:** Proxmox VE, LXC and VM isolation, Komodo, Prometheus, Grafana, Tailscale, zero-trust remote access.

EDUCATION

4/2021 - 3/2024	Bachelor of Computer Applications - Jain (Deemed-To-Be-University) Scored CGPA - 8.7	Undergraduate
4/2018 - 3/2021	PUC(SEBA) - Presidency University Scored 87.6%	PUC

WORK EXPERIENCE

10/2026 - Present	Lead Security Engineer	Cygne Noir Cyber
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- Hold sign-off on security architecture, tooling selection, and risk acceptance as lead security researcher and advisor for the managed security platform.
- Own the security research track: evaluate emerging tooling and detection approaches, prototype and benchmark them before they reach client deployments, and decide what ships.
- Run applied research on securing agentic pipelines and on LLM-assisted triage and enrichment, including the agent and prompt threat surface.
- Own the SOC 2 readiness program across all deployed environments: control mapping, gap remediation, and evaluation of Drata for compliance automation.
- Own cloud security across AWS and Azure client deployments, covering the per-client account isolation model, IAM boundaries, Cloudflare security and DNS, and ongoing posture management.
- Own CI/CD and production DevSecOps end to end: which gates run (Semgrep, Gitleaks, Trivy, Checkov, Syft), at which pipeline stage they run, and what blocks a merge or a deploy.
- Hire engineers, manage freelancers, and advise the founder on team performance, promotions, and role changes.
- Trained 7 security trainees on the platform's scanner output, DefectDojo triage workflow, and L2 SOC escalation path.

- Designed the security architecture for the managed platform: one EC2 instance per client in a separate AWS account, data locality as a hard rule so no scan input leaves the client deployment, Twingate zero-trust access, and an RBAC management API. The isolation model removes cross-tenant data exposure as a class of risk.
- Ran the scanner fleet across client deployments: Trivy, Prowler, Checkov, ScoutSuite, Semgrep, Gitleaks, and Kubescape.
- Built the findings pipeline on DefectDojo as the hub, with bidirectional JIRA integration, SARIF normalization, and LLM-assisted triage enrichment, including tracing CVE misattribution on shared-CDN infrastructure to cut false positives.
- Prioritized remediation with EPSS, CISA KEV, and SSVC, which moved known-exploited findings ahead of high-CVSS findings with no evidence of exploitation.
- Deployed Wazuh across client environments, integrated EDR and SIEM sources, and ran human-in-the-loop L2 SOC coverage with customer-facing incident support.
- Hardened production to zero trust: moved management and administrative interfaces off the public internet behind Twingate and removed standing direct SSH access to client instances.
- Owned AWS and Azure infrastructure and spend: ran a cost simulation at million-user scale, found S3 and network inefficiencies, and implemented optimizations projected to cut cloud costs by 90%.
- Designed security architecture for clients under EU regulatory regimes (DORA, NIS2, GDPR).

Engineering and delivery leadership

- Led a two-person application and API penetration testing function: set direction and methodology, ran white-box and black-box engagements, performed manual and automated code review, and wrote and signed off the client-facing reports.
- Built and shipped the backend for a production dating application as sole backend developer and technical lead (FastAPI, Supabase, Redis, AWS), designed to scale horizontally toward millions of users, in a 5-person product team of 3 frontend engineers, 1 backend engineer, and 1 QA engineer, delegating tasks, reviewing code, and setting development pace across the 3 frontend engineers.
- Designed the application security architecture for that product: autonomous threat detection, monitoring, and an automated ban system for real-time abuse response, plus the architecture plans and implementation roadmaps for later phases.
- Built agentic workflows and modular control pipelines for security automation and threat analysis, and contributed research on E2EE systems and secure agentic AI.
- Ran delivery end to end: sprint management, product-lifecycle planning, the client admin command center, and product and application security testing across 6+ research initiatives, of which 4 reached MVP and 1 is live in production.
- Managed clients directly through requirement-gathering sessions, change requests, and delivery communication.
- Spoke at university sessions and ran internal training for interns and employees.

- Built and shipped the compliance and operations software the business runs on, working remotely alongside the primary role.
- Provisioned and deployed those systems on on-premise hardware, including networking and edge deployment.
- Ran the self-hosted infrastructure on Proxmox VE with per-service LXC and VM isolation and secure remote access.
- Kept compliance-critical services available through monitoring, patching, and maintenance.

- Worked as the AI and backend engineer for one client across the longest of the four engagements, building and shipping Python and FastAPI services with LLM-backed processing.
- Researched and prototyped insurance filing automation for a second client.
- Ran online security enumeration work for a third client, the single security engagement of the four.
- Built a restaurant point-of-sale system for a fourth client and deployed the full backend on local hardware, including the networking adjustments and edge deployment.
- Built a design studio site for a designer selling third-party products on Amazon, kept at very low hosting cost.

- Ran cloud security assessments across AWS and Azure environments, focused on misconfiguration and exposure risk.
- Built and deployed security-focused Terraform environments for assessment and research workloads.
- Researched SBOM and supply-chain security tooling, Linux initialization behavior, and AWS security rules.
- Wrote Python security automation and red-team tooling, and supported vulnerability assessment, reporting, and remediation validation.

PROJECTS

AI security and LLM research

- **GPT-Vuln-Analyzer** (599 stars) - LLM-assisted vulnerability analysis across multiple model backends. *Python, LLMs, security analysis.*
- **HackBot** (363 stars) - agentic LLM assistant for security analysis and scan interpretation. *Python, LLMs, assistant systems.*
- **CVE-LLM-Dataset** (69 stars) - dataset design for training and evaluating LLMs on vulnerability intelligence. *Dataset, LLM research, CVE analysis.*

Security tooling and analysis

- **CloudG** - multi-cloud infrastructure mapping and audit CLI for AWS, Azure, and GCP: topology graphs with reachability and attack-path analysis, orchestrating Prowler, Checkov, Trivy, and ScoutSuite, with deduplicated findings scored against 28 compliance frameworks. *Python, cloud security, AWS, Azure, GCP.*
- **QuadraInspect** (349 stars) - automated Android APK security analysis framework. *Python, mobile security, reverse engineering.*
- **Nmap-API** (83 stars) - REST API wrapper around Nmap, and the basis for the published paper on API-driven network scanning. *Python, API development, network security.*
- **Startup-SBOM** - Linux boot and package analysis for startup and persistence vectors. *Python, Linux, SBOM.*

Systems and infrastructure

- **PICOTTY** (103 stars) - networked serial console system for headless fleet management, built on CircuitPython on Raspberry Pi Pico nodes with W5100S Ethernet, with over-the-air firmware updates, rollback protection, and canary rollout. Featured by Hackaday and CNX Software. *CircuitPython, Python, embedded.*
- **Komo.do-Hub** - mobile interface for the Komodo container platform. *Java, Node.js, React Native, React.*

Offensive security internals (background)

- **WinFiHack** (82 stars) - Windows wireless stack internals and native networking library automation. *Python, WiFi, Windows internals.*
- **C2C-Server** - command-and-control server demonstrating real attack communication patterns. *Python, C2, red team.*

HOMELAB

- Run a self-hosted Proxmox VE cluster with per-service LXC and VM isolation and Kubernetes workloads on custom hardware.
- Monitor the estate with Prometheus and Grafana, and orchestrate container deployments with Komodo.
- Reach every service over a Tailscale mesh with no public ingress, which is the same zero-trust pattern used in production.

CERTIFICATIONS

Valid 6-2023 to 6-2027	Certified Ethical Hacker - V12 Credential ID ECC7349261508.	EC-Council
Valid 6-2023 to 6-2027	Certified Network Defender Credential ID ECC9327805461.	EC-Council

PUBLICATIONS

2026	Hybrid Quantum-Classical Neural Network Incorporating Attention Mechanisms for Anomaly Detection A hybrid quantum-classical architecture with attention mechanisms for anomaly detection. My contribution was the computer science and systems side: architecture, implementation, and evaluation of the detection pipeline.	Springer CCIS
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2025	ML-Driven Secure Communication for Next-Generation 6G Networks A machine learning approach to secure communication in 6G networks, covering how ML can improve security and reliability in high-speed network infrastructure. Published in 6G Cyber Security Resilience: Trends and Challenges.	Springer
2024	Using Autoencoder-Driven Machine Learning for Advanced Cybersecurity Malware Detection Autoencoder-based models for malware detection on the Ember dataset, and how unsupervised feature learning finds anomalous behavior in executables. FMDB Transactions on Sustainable Intelligent Networks, Vol.1 No.4.	FMDB
2024	Docker Based Decentralized Vulnerability Assessment A decentralized vulnerability assessment framework built on Docker containers: distributed scanning, scaling, and AI-assisted port scanning and analysis. FMDB Transactions on Sustainable Intelligent Networks, Vol.1 No.4.	FMDB
2022	AI Based Enumeration and Exploit Suggester Using AI models in cybersecurity workflows to automate reconnaissance, vulnerability enumeration, and exploit suggestion, and to prioritize likely attack vectors. JETIR Vol.9 Issue 6.	JETIR
Year TBC	API-Based Network Scanning An API-driven approach to network vulnerability scanning, based on the Nmap-API project. Covers stability, usability, and scaling through structured resource management and virtualized client interactions.	Paper Link
Year TBC	AI in Action: Exploiting the Nexus of Cybersecurity AI in modern security environments, with a focus on integrating AI-driven analysis into CI/CD pipelines for security automation, threat detection, and response.	Paper Link

HONORS

Competition	Certificate of Merit Won first place in a research presentation competition at BGS College of Engineering and Technology.	BGS
Honor	Certificate of Appreciation Awarded by Jain (Deemed-to-be University) for speaking at the IRDCSTEM-2023 post-conference learning session.	JAIN